

1.2 FEATURES AND TECHNICAL DATA, HPRS

Functional Principle

LVDT stands for Linear Variable Differential Transformer. The LVDT is an analog and non-contact sensor with one primary and two secondary coils. An electrical signal is generated by the linear movement of the encoder rod. Attached to the encoder rod is the ferromagnetic core. The core induces a voltage into the secondary coils

Signal conditioning

An integrated signal conditioner converts the induced voltage into a output signal 4-20 mA. The actual measuring length 15 -21 mm can be tuned by two potentiometers to desired output signal range, default is 4-20 mA

Advantages

- Absolute measuring principle (position always available)
- Non-contact measuring principle, maximum service life
- Very low linearity deviations
- High repeatability
- Very high resolution with internal signal conversion
- Insensitive to external interference
- Mechanically very robust against external influences
- Pressure tube in stainless steel

FEATURE	RANGE, CHARACTERISTICS
Type approval	Presafe 14 ATEX 5124X, CE 2460 IECEX PRE 14.0021X
Identification	II 2G Ex db IIB T6 Gb
Supply voltage	24 VDC SELV DC supply with max. 24 VDC +20%
Max Current	62 mA
External Fuse	100 mA
Temperature range	Ambient: -40°C ... +55°C; Storage > 24 months +5°C ...+25°C (antistatic bag)
Measuring principle	LVDT with integrated electronic
Output Signal	4-20 mA (12 mA +/- 8 mA)
Measuring stroke	21 mm ($\pm$ 10,5 mm)
IP protection	IP67
Measuring principle	LVDT with integrated electronic
Linearity	$\pm$ 0,5 % F.S.
Repeatbility	$\pm$ 0,1 % F.S.
Output load	< 500 Ohm
Temperature coefficient	< $\pm$ 0,05 % F.S. / °C
Adjustments	Offset and gain by 10 turns potentiometer
Encapsulation	Sensor housing and cable adaptor: Stainless steel Valve adaptors: Steel with Corridur surface treatment
Connection to application	3-wire +PE 0,5-0,75 mm ² , flying lead, 0,5; 1,0; 3,0; 10,0 m; with/without shield
EX HPRS User Manual	Servi doc: 1503132

	RANGE, CHARACTERISTICS
Housing Material	Sensor housing and Cable adaptor: AISI 316 L Valve adaptor Steel EN 10027-2, 1.0338
Marking	Laser engraved into sensor housing
Main dimensions	Sensor housing with horizontal cable adaptor: l = 68 ¹ ; h = 75; w = 44 [mm] Sensor housing with vertical cable adaptor: l = 68 ¹ ; h = 92 ¹ ; w = 44 [mm] ¹) Minimum bending radius flying lead not included Valve adaptor dimensions not included
Earthing (located at cable adaptor)	M5 x 10 screw in A4 70/80 including spring washer in A4 DIN 128 A4/70
Access for adjustments	M4 x 8 A4/70 with CU seal ring Ø4 x 8 (2 sets)
Fastening bolts	Size 3: Cylinder Head Cap Screw, M5 x 115 A4 70/80 Size 5/7: Cylinder Head Cap Screw, M5 x 68 A4 70/80
Cable Gland	Shielded cable: Eaton Crouse Hinds ADE 4F, type 5 or 6. M16 Non shielded cable: Eaton Crouse Hinds ADE 1F2, type 4 or 5. M16